

RACHIT KUMAR TIWARI

B.Tech CSE · MCKV Institute of Engineering · Expected May 2028 · Kolkata, India

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GITHUB LINKEDIN PORTFOLIO

CAREER OBJECTIVE

3rd-year B.Tech CSE student who has shipped **10+ production-grade applications** spanning AI-integrated full-stack systems, real-time computer vision pipelines, and clinical-grade safety platforms. Awarded **2 hackathon wins and 3 national finalist positions** for building tools that solve real problems — from pharmaceutical waste reduction to desktop privacy for enterprise environments. Looking to bring hands-on experience with Next.js, Python, FastAPI, and LLM orchestration into a high-impact engineering team.

EDUCATION

- MCKV Institute of Engineering, West Bengal

B.Tech in Computer Science & Engineering

Coursework: Data Structures & Algorithms, OOP, DBMS, Computer Networks, OS, Software Engineering, Web Technologies

2024 – 2028 (Expected May 2028)

Cumulative YGPA: 8.12/10.0
- St. Aloysius Orphanage and Day School, Kolkata

Class XII (ISC)

2022
- St. Aloysius Orphanage and Day School, Kolkata

Class X (ICSE)

2020

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C
Frameworks: Next.js, React, Node.js, FastAPI, Electron, Tailwind CSS
AI/ML & CV: OpenAI API, Gemini API, Mistral AI, OpenCV, MediaPipe, Face Recognition, RAG Pipelines
Cloud & Tools: Firebase, Supabase, MongoDB, Docker, Git, Postman, Figma, Vercel

FEATURED PROJECTS

- MedCheck

React, Vite, FastAPI, Python, Mistral AI, Docker

AI-Powered Clinical Drug Interaction Platform

– Built clinical drug-interaction platform with **17 deterministic pharmacology rules** + Mistral AI fallback for unstructured FDA labels; multi-tier severity classification reduced false positives.

– Architected FastAPI backend with composite GI risk scoring, JWT/bcrypt auth; **containerized with multi-stage Docker**. Delivered responsive React/Vite frontend with real-time search and clipboard-export summaries.
- PrivSight

Electron, React, TypeScript, Python, OpenCV

Real-Time Desktop Privacy System via Facial Recognition

– Engineered desktop privacy guard that **auto-blurs screens when unauthorized faces detected** via webcam using real-time facial recognition with OpenCV.

– Achieved **~30 FPS face detection with sub-100ms latency** via optimized WebSocket bridge between Electron frontend and Python backend; **100% on-device** — zero cloud dependency.
- Vitamend

Next.js 14, TypeScript, MongoDB, Gemini Vision

AI-Driven Medicine Donation & Redistribution Platform

– Developed secure medicine donation platform tackling pharmaceutical waste; built **Gemini 1.5 Flash Vision OCR pipeline** extracting medicine details from label images with confidence scoring.

– Implemented QR-based chain-of-custody tracking linking donors, verifiers, and recipients. **Winner:** Innovation Ideas 2024-25 & MCKVIE Research Symposium | **Finalist:** Cosmohack Top 10, HACKVSIT Top 20.
- DocEasy

Next.js 16, TypeScript, Supabase, Tailwind CSS

Document Management & PDF Processing Platform

– Built document platform with industrial PDF compression and **O(1) memory complexity via Web Streams API** for payloads up to 250MB; hybrid client-side WebAssembly + server-side streaming architecture.

– Implemented Resume Parsing, ATS scoring, and PDF summarization backed by Supabase Auth (OAuth) and Row-Level Security PostgreSQL.

OTHER PROJECTS

Kred.ai (Next.js 16, Gemini API, Serper API) — RAG fact-checking pipeline with real-time web search and confidence scoring.
Vocalis OS (Next.js 16, Firebase, OpenAI API) — Voice-first AI OS with Web Speech API and autonomous agents.
Kolkata Metro Route Planner (Next.js 15, TypeScript) — Dijkstra’s algorithm across 6 metro lines with peak-hour multipliers.
Vitals.io (React, TypeScript, Python, D3.js) — SaaS unit economics analyzer with fuzzy CSV detection.
Air-Mouse (Python, OpenCV, MediaPipe) — Touchless HCI with Kalman filtering and 21-point hand landmarks.
OptraFi (React, TypeScript, Firebase, OpenAI API) — AI personal finance platform with real-time tracking.
WELCOMETOMYCITY (Next.js 15, TypeScript, MapLibre) — Map-first travel platform with emotional vibe mapping.
Full source code and demos available at github.com/rachts

ACHIEVEMENTS & LEADERSHIP

Winner: Innovation Ideas 2024-25 (MCKVIE) & MCKVIE Research Symposium (Vitamend)
Finalist: Hacklabify V1.0 Top 8 (Vitals.io) | Cosmohack Top 10 (Vitamend) | HACKVSIT Top 20 (Vitamend)

CERTIFICATIONS

HackerRank: SQL (Advanced, Intermediate, Basic), JavaScript (Intermediate), Node.js (Intermediate), REST API (Intermediate), Problem Solving (Intermediate), Java (Basic), Python (Basic), CSS (Basic)